Finite-population correction (FPC)

⚠ PROVISIONAL / LEGACY-SUPPORTED BRIDGE

1 RECOGNISE IT

- A finite population size N is supplied.
- Sampling is without replacement.
- Sampling fraction n/N is more than 0.05.

2 DECISION / METHOD

1. Confirm finite N and without replacement.
2. Compute ordinary SE.
3. Compute FPC.
4. Multiply SE by FPC before standardising.
5. If with replacement or n/N<=0.05, keep ordinary SE.

3 FORMULA

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FPC=sqrt((N-n)/(N-1))
adjusted SE = ordinary SE * FPC
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4 FORMULA SHEET VS WHAT I KNOW

Status: C7.52-C7.58 retain the canonical master's provisional scope safeguard. Bridge: retained as a sourced bridge, especially into Chapter 8; not silently promoted as current Chapter 7 core.

5 EXCEL ROUTE

- =SQRT((N-n)/(N-1))
- adjusted mean SE: =sigma/SQRT(n)*FPC
- adjusted p SE: theoretical SE times FPC.

6 TINY WORKED EXAMPLE

N=500,n=50 without replacement: FPC=SQRT(450/499)=0.9496. Multiply the ordinary SE by 0.9496.

7 TOP EXAM TRAPS

- Applying FPC merely because N is stated.
- Applying it when sampling with replacement.
- Omitting the square root.
- Failing to inspect n/N.

8 SAY ALOUD

Use FPC only for material sampling without replacement from a finite population: inspect replacement and n/N first.

9 EXIT CHECK

N=1000,n=20 without replacement: does the 5% rule trigger FPC?

ONE-GLANCE VISUAL ROUTE

